Operations Research III: Theory

Quiz for Week 5 (Convex Analysis)

Instructor: Ling-Chieh Kung
Department of Information Management
National Taiwan University

1. For each of the following sets, select all that are convex:
   (a) $S = \{(x_1, x_2) \in \mathbb{R}^2 | x_1^2 + x_2^2 \leq 4, x_1 \geq 0, x_2 \leq 0\}$.
   (b) $S = \{(x_1, x_2) \in \mathbb{Z}^2 | x_1^2 + x_2^2 \leq 4, x_1 \geq 0, x_2 \leq 0\}$.
   (c) $S = \{x \in \mathbb{R}^n | \sum_{i=1}^n x_i^2 \leq 4, x_1 \geq 0, x_2 \leq 0\}$.
   (d) $S = \{x \in \mathbb{R}^n | x_1 + x_2 = 10(x_3 + x_4)\}$, where $n \geq 4$.
   (e) $S = \{x \in \mathbb{R}^n | \max_{i=1,\ldots,n}\{x_i\} = 10 \min_{i=1,\ldots,n}\{x_i\}\}$.

2. For each of the following functions, select all that are convex over the given region:
   (a) $f(x) = 2x^3 - x^2 - 2x + 1$ for $x \in \mathbb{R}$.
   (b) $f(x) = \begin{cases} -x & \text{if } x < 1 \\ -1 & \text{if } x \geq 1 \end{cases}$ for $x \in \mathbb{R}$.
   (c) $f(x) = x_1^2 - 4x_1x_2 + 3x_2^2 + 3x_1 + 4x_2$ for $x \in \mathbb{R}^2$.
   (d) $f(x) = \sum_{i=1}^n (x_i - a)^2$, where $a > 0$ is a given parameter, for $x \in \mathbb{R}^n$.
   (e) $f(\alpha, \beta) = \sum_{i=1}^n (\alpha + \beta x_i - y_i)^2$, where $x_i \in \mathbb{R}$ and $y_i \in \mathbb{R}$ are given parameters, $i = 1, ..., n$, for $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}$.

3. For the nonlinear program

$$\begin{aligned} \min \quad & 2x^3 - x^2 - 2x + 1 \\ \text{s.t.} \quad & x \geq -1, \end{aligned}$$

   Which of the following statements is correct?
   (a) This is a convex program.
   (b) There are two local minimizers.
   (c) The unique global minimizer is a boundary point.
   (d) This program is unbounded.
   (e) None of the above.

4. For each of the following matrices, select all that are positive semi-definite:
   (a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
   (b) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$.
   (c) $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$.
   (d) $A = \begin{bmatrix} 1 & 2 & 3 & \cdots & n \\ 0 & 2 & 3 & \cdots & n \\ 0 & 0 & 3 & \cdots & n \\ \vdots & \vdots & \vdots & \ddots & \\ 0 & 0 & 0 & \cdots & n \end{bmatrix}$ for some $n \geq 3$.

   (e) $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.

5. For the following statements, select all that are incorrect:
   (a) An optimal solution to a linear program must be a boundary point.
   (b) An optimal solution to a linear program must be an extreme point.
   (c) An optimal solution to a nonlinear program can be an interior point.
   (d) An optimal solution to a nonlinear program must be an interior point.
   (e) A global optimal solution to a nonlinear program must be a local optimal solution.